

Kasra Ahmadi

ML Research Scientist & Engineer · LLM Systems · ML Security & Privacy

kasra.research@gmail.com | [Google Scholar](https://scholar.google.com/citations?user=KasraAhmadi) | github.com/KasraAhmadi | [linkedin.com/in/kasra-ahmadi](https://www.linkedin.com/in/kasra-ahmadi) | Sunnyvale, CA

PROFESSIONAL SUMMARY

Ph.D. candidate in Computer Science and Engineering and ML researcher with **10+ peer-reviewed papers**. At RealmLabs, designed a compact frozen-backbone **PII span-localization** model that beat OpenAI's dedicated 1.5B Privacy Filter under a shared leakage-free evaluation (**0.837 vs. 0.654 mean micro-F1**), then shipped it into a shared vLLM serving stack. Received the **Best Paper Award** at the 2025 IEEE Security and Privacy Workshops for differentially private federated learning. A researcher who builds the system behind the paper — from novel model design to production LLM infrastructure.

WORK EXPERIENCE

RealmLabs

May 2026–Aug 2026

ML Research Engineer Intern

Sunnyvale, CA

- Designed and trained an exact PII span-localization model — a 634K-parameter head (0.007% of the backbone) over a fully frozen Qwen3.5-9B — that beat OpenAI's dedicated 1.5B Privacy Filter under one shared leakage-free harness: **0.837 vs. 0.654 mean micro-F1** across ai4privacy 200K/300K/400K.
- Removed the causal-decoder limitation without modifying the backbone, using a learned ELMo-style mixture over all 32 hidden-state layers, an external bidirectional attention read-out, and calibrated constrained Viterbi decoding.
- Consolidated generation, PII, sparse-autoencoders, and role-aware probes into one FastAPI service over a shared frozen backbone, backed by a specialized **vLLM fork** exposing hidden states during prefill and decode with continuous batching and prefix caching.
- Benchmarked Qwen3.5-9B on an A100-80GB: generation throughput scaled **64 → 267 tokens/s** (concurrency 1 → 200) with zero errors, and validated 97.6% on-GPU SAE parity against the Hugging Face path.

University of South Florida

Jan 2022–Present

Ph.D. Researcher

Tampa, FL

- Designed a human-in-the-loop, differentially private federated-learning framework for fine-tuning BERT-class language models on memory-constrained devices — **Best Paper Award**, 2025 IEEE Security and Privacy Workshops (HMI-SA).
- Built **MAED**, an algorithm-level runtime error-detection framework using mathematical identities to catch fault-injection attacks in DNN activation functions.
- Studied white-box, gradient-based adversarial attacks that perturb vector embeddings to degrade RAG retrieval and downstream answer quality.
- Developed side-channel and fault-detection countermeasures for NIST post-quantum standards (CRYSTALS-Kyber, Dilithium); authored 10+ peer-reviewed papers under an NSF-funded program.

TD SYNnex

May 2025–Aug 2025

AI Engineer Intern

Clearwater, FL

- Designed cloud-native LLM, RAG, and multi-agent workflows for Jarvis, a multilingual hospitality assistant on NVIDIA AI Enterprise — enterprise-file retrieval, NIM services, agent memory, tool calls, and NeMo Guardrails for input/output safety.

PaarLift

Jan 2018–Apr 2020

ML Engineer

Tehran, Iran

- Built ML-driven elevator dispatch (logistic regression, KNN) that cut passenger wait times **27% across 100+ commercial buildings**; Raspberry Pi / CAN-bus IoT ingestion and Apache Airflow ETL into an OLAP store.

SELECTED PROJECTS

1. **job-eval** *Founder and Builder · 1K active users* GitHub

Launched an explainable resume-to-job matcher powered by a purpose-trained 156M-parameter ModernBERT cross-encoder with four scoring heads; model-native sparse-attention evidence replaces a second generative model.

ModernBERT · PyTorch

2. **PII-360** *Open-source Chrome extension* GitHub

Detects and masks multilingual PII in images, text, and PDFs with compact ONNX models that run entirely in the browser, so source material never leaves the device.

ONNX · React · TypeScript

3. **Gradient-Based RAG Attack** *Open-source research* GitHub

White-box, gradient-guided perturbations at the embedding layer that break retrieval and weaken downstream RAG answers.

PyTorch · embedding-space attacks

EDUCATION

Ph.D. in Computer Science and Engineering *University of South Florida* Jan 2022 – Oct 2026 (expected)

ML Security, Privacy-Preserving ML, Side-Channel and Fault Analysis, Post-Quantum Cryptography

Master of Information Technology Engineering *Amirkabir University of Technology* 2018 – 2022

SELECTED PUBLICATIONS

[Google Scholar](#)

All papers first-authored.

1. **An Interactive Framework for Implementing Privacy-Preserving Federated Learning: Experiments on Large Language Models**

K. Ahmadi et al. · IEEE Security and Privacy Workshops, 2025 · **Best Paper Award**

2. **PII Span Localization with a Frozen Autoregressive LLM**

K. Ahmadi et al. · RealmLabs manuscript, 2026

3. **MAED: Mathematical Activation Error Detection for Mitigating Physical Fault Attacks in DNN Inference**

K. Ahmadi et al. · Manuscript in submission

4. **Efficient Algorithm-Level Error Detection for the Number-Theoretic Transform Assessed on FPGAs**

K. Ahmadi et al. · ACM Transactions on Embedded Computing Systems, 2025

TECHNICAL SKILLS

ML and LLMs: PyTorch, TensorFlow, Hugging Face, ModernBERT, Qwen3.5, Llama 3.1, scikit-learn, LoRA, DP-SGD, Flower

LLM Systems: vLLM, hidden-state extraction, sparse autoencoders, model probes, RAG, FastAPI, ONNX, Docker

ML Security and Privacy: PII detection, adversarial ML, differential privacy, federated learning, fault injection, side-channel analysis

Programming and Cloud: Python, C++, C, Linux/Bash, AWS Lambda, Glue, S3